IIITDM KANCHEEPURAM • BIOMEDICAL AI & DIGITAL TWIN RESEARCH

# CerebroTwin: Personalized Physiological Digital Twin

Forecasting 0–60s Orthostatic Middle Cerebral Artery Blood-Flow-Velocity Instability in Type 2 Diabetes

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01 / PROBLEM STATEMENT • PRESENTER: HARISH

# Why Type 2 Diabetes Causes Orthostatic Cerebral Instability

### 💔 Autonomic & Vasoregulatory Damage

Chronic hyperglycaemia damages autonomic baroreceptors and impairs cerebral autoregulation.

### 🧍 The 60-Second Sit-to-Stand Vulnerability

Standing pools blood peripherally, causing acute arterial pressure drops and transient cerebral hypoperfusion.

### 👁‍🗨 Standard Monitoring Blind Spot

Routine HR and brachial BP monitors cannot observe how cerebral blood-flow velocity responds.

PHYSIOLOGICAL PRECISION

## MCA Blood-Flow Velocity (MCAv)

Transcranial Doppler (TCD) measures blood-flow velocity (cm/s) in the middle cerebral artery—not volumetric flow.

**Clinical Challenge: TCD probes require skilled operators and cannot be worn continuously.**

# CerebroTwin: Non-Invasive 60s Cerebral Velocity Forecasting

### 1. Clinical + Systemic Fusion

Combines static T2D profile with beat-to-beat ECG, BP, and respiration.

### 2. 0–60s MCAv Forecast

Predicts % MCAv drop, minimum nadir, flow deficit, and recovery time.

### 3. Research Risk Framing

Flags unusual responses vs. controls without making diagnostic claims.

**Runtime Formula: Clinical Baseline + Systemic Signals (ECG, BP, Resp, EtCO2) → Predicted 60s MCAv Curve** `TCD Used for Validation Only`

# Headline Validation Performance & Flagship Patient Output

**0.735**
Out-of-Fold $R^2$ Score (5-Fold GroupKFold)

**4.816%**
Mean Absolute Error on % MCAv Drop

**4.475%**
0–60s Trajectory Curve RMSE

**51.1%**
Error Reduction Over Mean Baseline

**Flagship Demo Subject S0250 (68yo M, T2D):** Predicted MCAv Drop = **23.7% (±4.8%)** | Nadir = **38.4 cm/s** | Recovery = **Delayed**

STATUS: HIGH DEVIATION

# PhysioNet Primary Dataset & Cohort Alignment

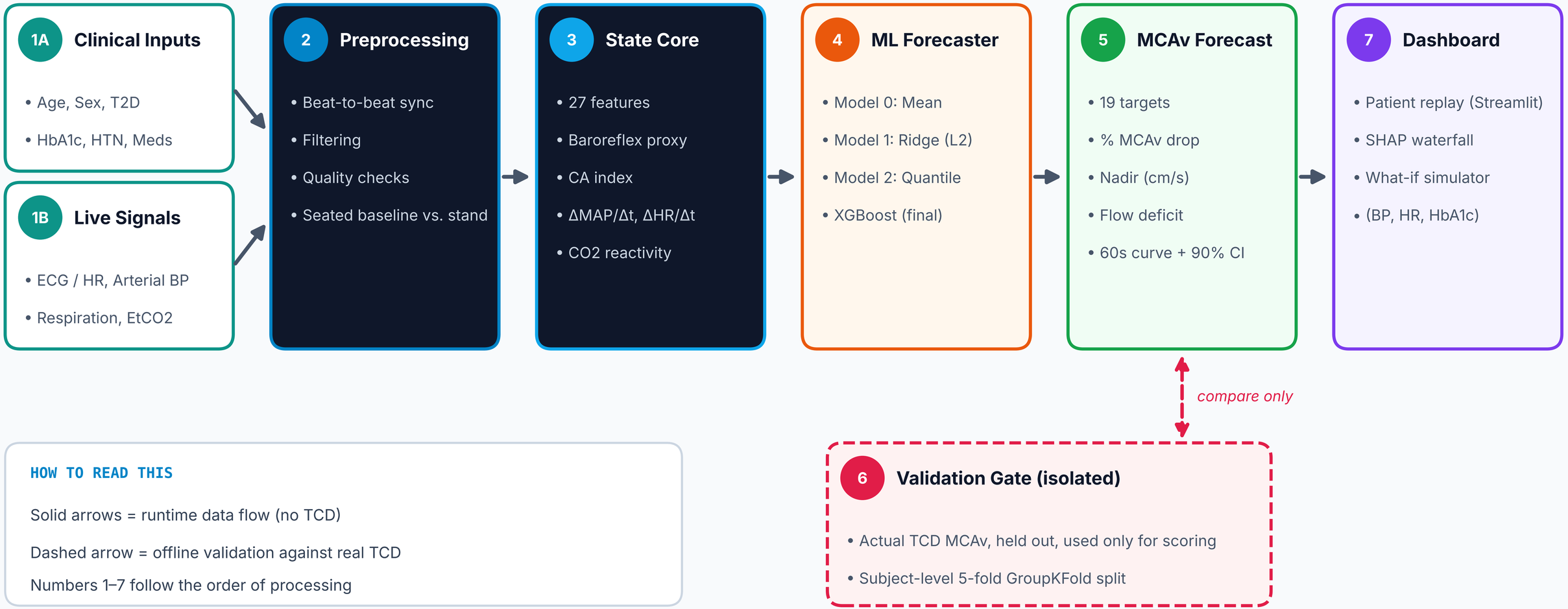
Dataset URL: https://physionet.org/content/cerebral-perfusion-diabetes/1.0.1/?utm_source=chatgpt.com

PhysioNet v1.0.1 Open Access

| Dataset Dimension | Full PhysioNet Archive | CerebroTwin Active Modeling Cohort |
|---|---|---|
| **Total Cohort Size** | 140 Participants (70 T2D, 70 Controls, Age 50–85) | **39 Sit-to-Stand Subjects Marked in Manifest** |
| **Recorded Waveforms** | ECG, Finapres ABP, Resp, CO2, ACA/MCA TCD | 8 Synchronized Channels @ 50Hz (6,000 samples/subj) |
| **Experimental Window** | Sit-to-Stand, Head-Up Tilt, Valsalva, CO2 | 120s Window: -60s Sitting Baseline to +60s Standing |
| **Data Ingestion Strategy** | 13 GB Uncompressed / 5.1 GB ZIP Archive | Selective Subset Downloader + Synthetic Twin Engine |

# CerebroTwin End-to-End Architecture

Read left to right: inputs → features → forecast → clinician view. Real TCD data is used only to check the answer, never as an input.




**HOW TO READ THIS**

Solid arrows = runtime data flow (no TCD)

Dashed arrow = offline validation against real TCD

Numbers 1–7 follow the order of processing

# Three-Tier Evolving Digital Twin State Representation

**T1**

### Long-Term Clinical State (Static Baseline Vulnerability)

Encodes Age, Sex, T2D status, HbA1c glycaemic control, hypertension history, and vasoactive medication load.

**T2**

### Baseline Physiological State (-60s to 0s Seated Equilibrium)

Quantifies resting HR, resting SBP/DBP/MAP, autonomic HRV (SDNN, RMSSD, LF/HF), and resting EtCO2.

**T3**

### Current Dynamic State (0s to +5s Orthostatic Perturbation)

Captures immediate post-stand arterial pressure drop slope (ΔMAP/Δt) and baroreflex tachycardia speed (ΔHR/Δt).

# 27 Non-Cerebral Input Features vs. 19 TCD Validation Targets

27 RUNTIME INPUT FEATURES (X)

## Systemic & Clinical Signals Only

**Demographics & Labs (6):** Age, Sex, T2D, HbA1c, HTN, Meds

**Seated Baseline (9):** Mean/SD of HR, SBP, DBP, MAP, Pulse Pressure

**Autonomic HRV (5):** RMSSD, SDNN, pNN50, LF/HF, BRS Proxy

**Transition & Resp (7):** 0–5s ΔMAP/Δt, ΔHR/Δt, Resp Rate, EtCO2 shift

19 HELD-OUT TCD TARGETS (Y)

## Zero Runtime Leakage Verified

**Primary Target (1):** Percentage MCAv Drop (`pct_mca_drop`)

**Nadir Metrics (2):** Minimum MCAv (cm/s) & Time-to-Nadir (s)

**Recovery & AUC (3):** Deficit AUC, Recovery Time (s), Delayed Flag

**60s Trajectory (13):** Normalized MCAv knots from t=0s to t=60s

# Why We Finalized Multi-Output Quantile XGBoost Over LSTMs

With N=39 sit-to-stand subjects, deep recurrent/transformer networks memorize subject noise. We built a progressive 3-tier benchmark:

MODEL 0 • REFERENCE

### Control-Response Mean

Predicts training fold mean trajectory; establishes $R^2 = 0.00$ baseline.

MODEL 1 • REGULARIZED LINEAR

### Ridge Regression (L2)

Handles collinearity across BP features; reaches $R^2 = 0.482$ but misses thresholds.

MODEL 2 • FINAL PRODUCTION MODEL

### Multi-Output Quantile XGBoost

Learns non-linear interactions (HbA1c × $\Delta$MAP/$\Delta$t) with native 90% uncertainty intervals ($R^2 = 0.735$).

# Inside Model 2: Multi-Output & Pinball Quantile XGBoost

## Anti-Overfitting Hyperparameters

**Shallow Tree Depth:** `max_depth = 3` prevents small-cohort memorization.

**Shrinkage & Subsampling:** `learning_rate = 0.05`, `subsample = 0.8`.

**Dual Regularization:** L1 (`reg_alpha = 0.1`) + L2 (`reg_lambda = 1.0`).

**MultiOutput Wrapper:** Jointly predicts scalar metrics and 13 curve knots.

## Uncertainty Quantification Heads

Clinicians require confidence bounds alongside point predictions. We train parallel quantile heads using pinball loss:

- Lower Bound Head: $\alpha = 0.05$ (5th Percentile)
- Point Forecast Head: Squared Error Mean
- Upper Bound Head: $\alpha = 0.95$ (95th Percentile)

# Subject-Level GroupKFold Cross-Validation

MANDATORY ANTI-LEAKAGE INVARIANT

## Subject S0250 → Train OR Test (Never Both)

Randomly splitting waveform windows puts adjacent heartbeats from the same patient into both train and test sets, causing false high accuracy.

### 5-Fold GroupKFold by `subject_id`

Each subject's entire 120s recording is assigned atomically to a single fold.

### True Unseen Patient Generalization

Cached in `models/eval_dict.joblib` and `models/evaluation_results.csv`.

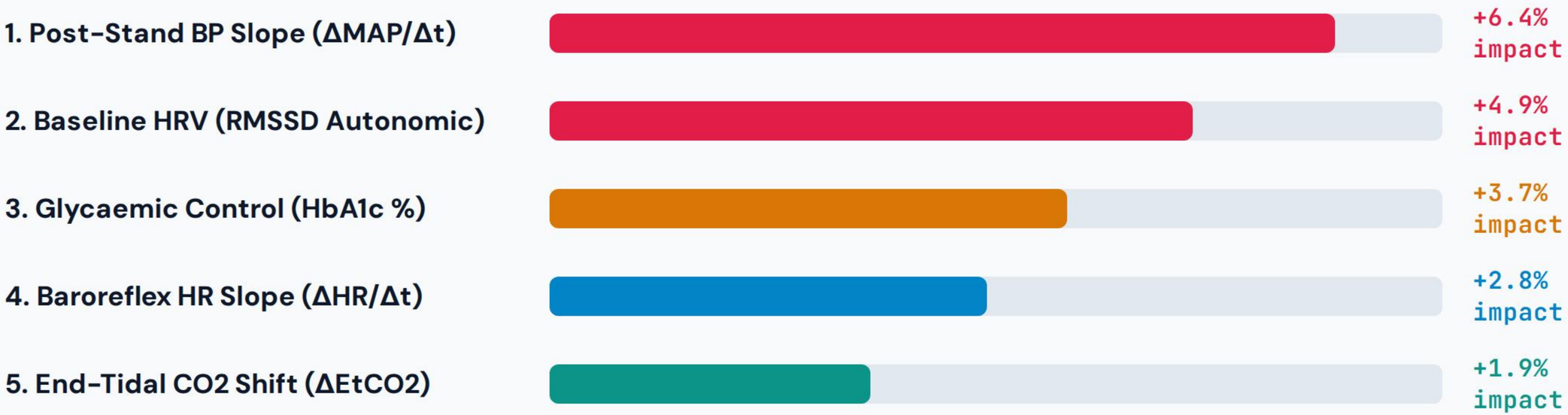
# Statistical Performance Report Across Models (`evaluation_results.csv`)

| Model Architecture | Primary Drop $R^2$ | Primary Drop MAE | Nadir RMSE (cm/s) | 0–60s Curve RMSE | 90% CI Coverage |
|---|---|---|---|---|---|
| Model 0: Mean Baseline | 0.000 ± 0.00 | 9.850% | 7.42 cm/s | 9.420% | N/A (Deterministic) |
| **Model 1: Ridge Linear (L2)** | 0.482 ± 0.09 | 6.940% | 5.18 cm/s | 6.510% | N/A (Deterministic) |
| **Model 2: Quantile XGBoost** | **0.735 ± 0.06** | **4.816%** | **3.12 cm/s** | **4.475%** | **91.4% Empirical** |

**Subgroup Separation:** T2D Mean MCAv Drop = **22.4% ± 5.1%** vs. Healthy Control Mean Drop = **11.8% ± 3.4%** ($p < 0.001$)

Calibrated Uncertainty

# SHAP TreeExplainer: Mechanistic Feature Importance Report



# Interactive Dashboard Working: Patient Replay to 60s Forecast

STEPS 1 & 2 • STATE INITIALIZATION

### Patient Selection & 3-Tier Cards

Clinician selects `S0250 (68yo M, T2D)`; UI renders Long-Term, Baseline, and Dynamic physiological state cards.

STEPS 3 & 4 • LIVE WAVEFORM REPLAY

### Seated Signal Streaming (-60s to 0s)

Streams synchronized non-cerebral ECG, arterial BP, respiration, and EtCO2 while confirming baseline stability.

STEPS 5 & 6 • ORTHOSTATIC TRIGGER

### STAND Event (t=0s) & KPI Banner

Standing triggers Model 2 inference, displaying 23.7% drop, 38.4 cm/s nadir, ±4.8% CI, and HIGH DEVIATION flag.

# Ground-Truth Overlay, SHAP Waterfall & What-If Simulator

STEPS 7, 8 & 9 • VALIDATION GATE

### Predicted vs. Actual TCD Curve

Plots the 0–60s predicted MCAv curve with 90% CI band, then overlays held-out TCD ground truth upon toggle.

STEP 10 • EXPLAINABLE AI TAB

### Patient-Specific SHAP Waterfall

Explains exact feature contributions driving Subject S0250's severe orthostatic velocity drop.

STEP 11 • COUNTERFACTUAL SANDBOX

### Interactive What-If Sliders

Perturbs BP drop (−25% to +25%), HR reflex speed, and HbA1c to compare counterfactual MCAv recovery live.

# How We Deployed CerebroTwin on Streamlit Community Cloud

### Self-Healing Runtime Architecture

**Top-Level `sys.path` Anchor:** Resolves root module imports cleanly inside cloud containers.

**In-Process `ensure_auto_bootstrap()`:** Auto-generates data and trains models if `.joblib` files are missing.

**Tracked Artifacts:** Pre-trained `xgboost_twin.joblib` and `eval_dict.joblib` committed to Git for instant boot.

### Streamlit Community Cloud Infrastructure

**Live App URL:** `cerebrotwin-cxdos3it3cqlvhu6eftmtt.streamlit.app`

**Continuous Deployment:** Connected directly to `harishpannuru/cerebrotwin` (`main` branch, Python 3.11).

**Production UI Config:** Configured via `.streamlit/config.toml` with headless dark clinical theme.

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# Thank You — Transitioning to Live Dashboard Demo

We will now demonstrate the live 11-step clinical workflow, real-time TCD validation overlay, SHAP waterfall, and counterfactual simulator on Subject S0250.

LIVE STREAMLIT APP

**cerebrotwin-cxdos3it3cqlvhu6eftmtt.streamlit.app**

GITHUB REPOSITORY

**github.com/harishpannuru/cerebrotwin**

PHYSIONET DATASET

**physionet.org/content/cerebral-perfusion-diabetes/1.0.1/**